



슈퍼컴퓨팅센터
GIST Supercomputing Center

V100 GPU 테스트 가이드

작성자: 유은호



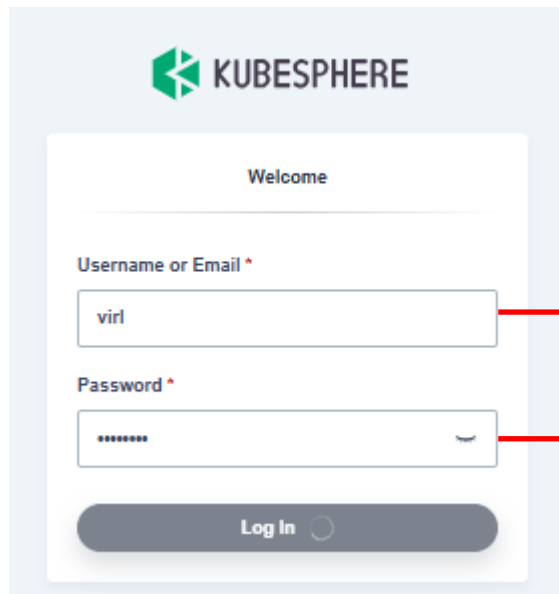
A large, bold, red stylized letter 'I' is centered in the upper half of the dark gray panel.

overview

접속 방법



브라우저 주소창 입력 http://210.125.70.71:30880



The image shows a login interface for KUBESPHERE. At the top is the KUBESPHERE logo. Below it is a 'Welcome' message. There are two input fields: 'Username or Email' with the text 'virl' and 'Password' with masked characters. A 'Log In' button is at the bottom. Red arrows point from the input fields to the labels 'ID :' and 'PW :' on the right.

계정 접속 정보

ID :

PW :



Change Password

You are logging in for the first time. Please change your password.

Password *

Your password must:

- ☒ Contain at least 1 uppercase letter and 1 lowercase letter.
- ☒ Contain at least 1 number.
- ☒ Contain at least 1 special character
(~!@#\$%^&*()-_+=\|[]{};:'",<.>/? or space).
- ☒ Contain at least 8 characters.

Submit

패스워드 변경

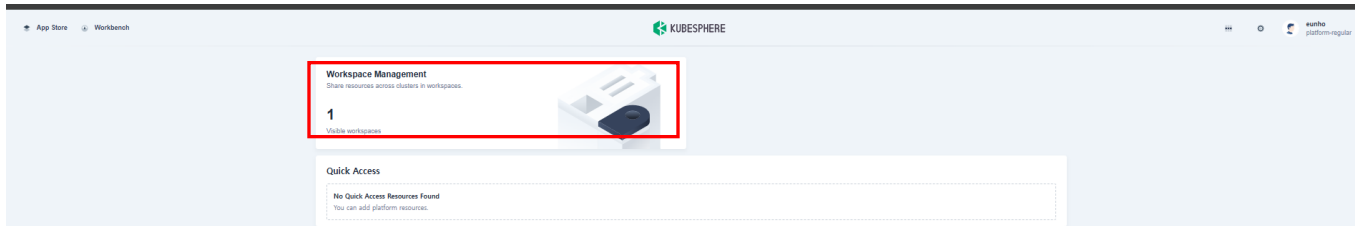
II

overview

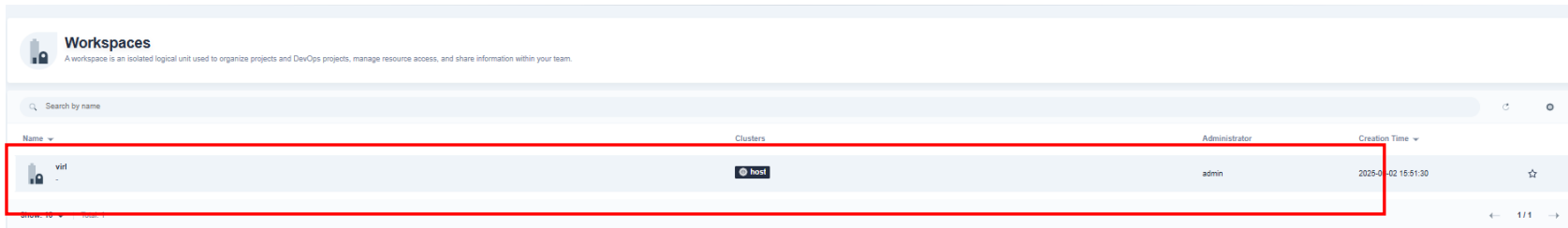
작업 공간 진입

1. 로그인

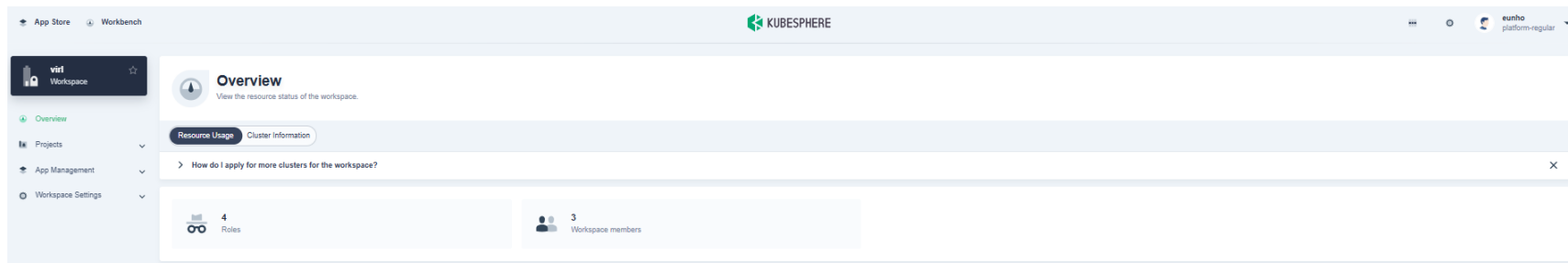
① 로그인 초기화면



② 'Workspace Management' 클릭

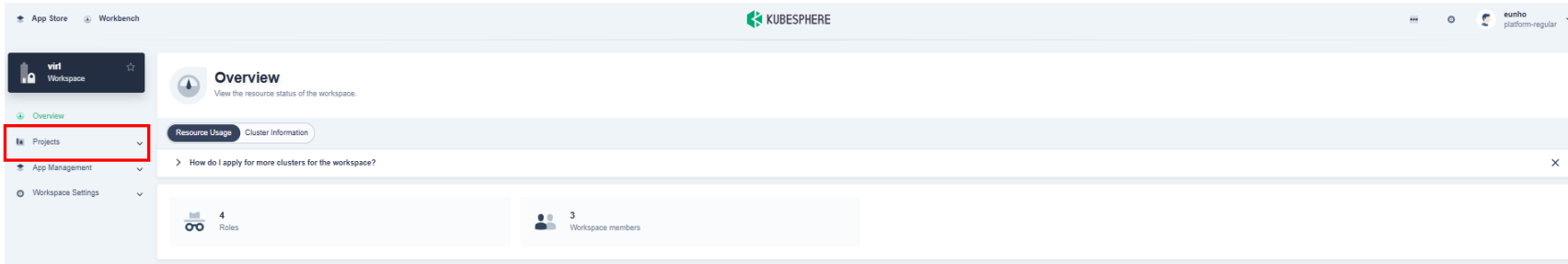


③ 'virl' 클릭

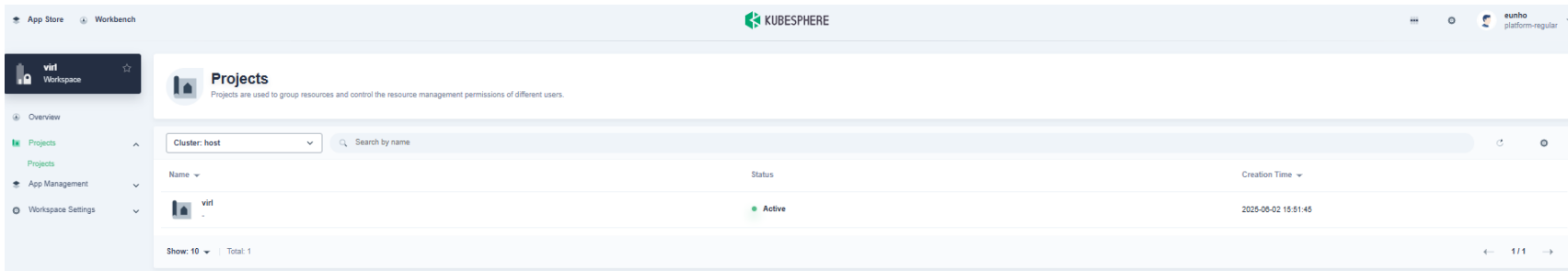


1. 로그인

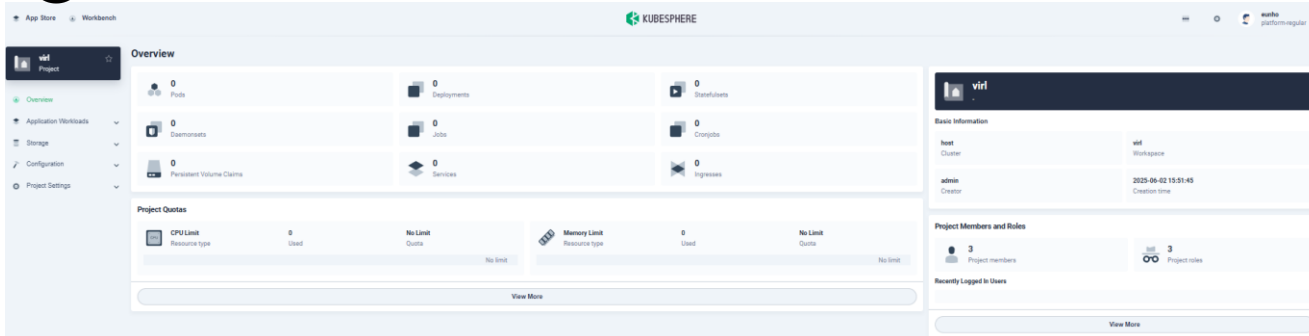
④ 'Projects' 클릭



⑤ 'Projects' > 'virl' 클릭



⑥ 할당된 리소스 확인



CPU, Memory: 제한 없음
PVC Limit : Host 머신 디스크 용량 1Ti 제한
PVC : Storage 용량 4Ti (4095Gi)
GPU : V100 24장

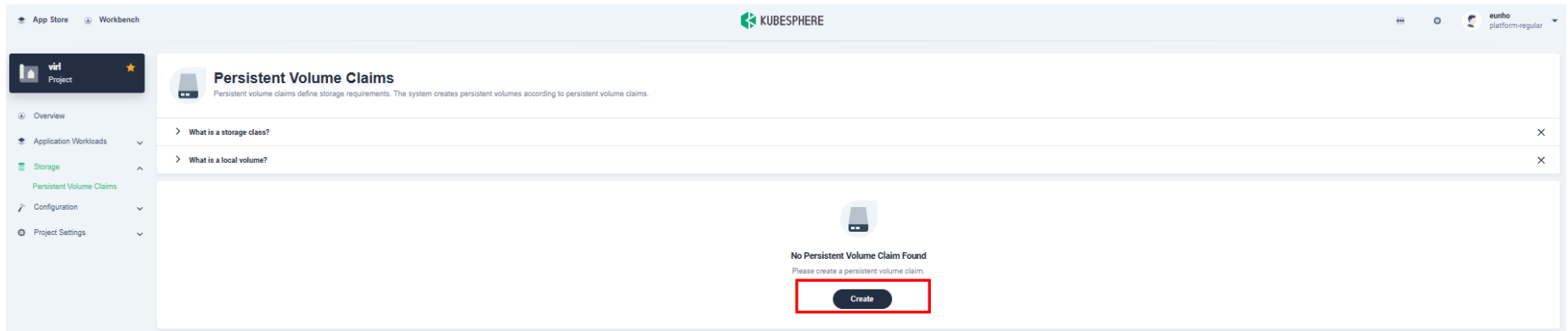
III

overview

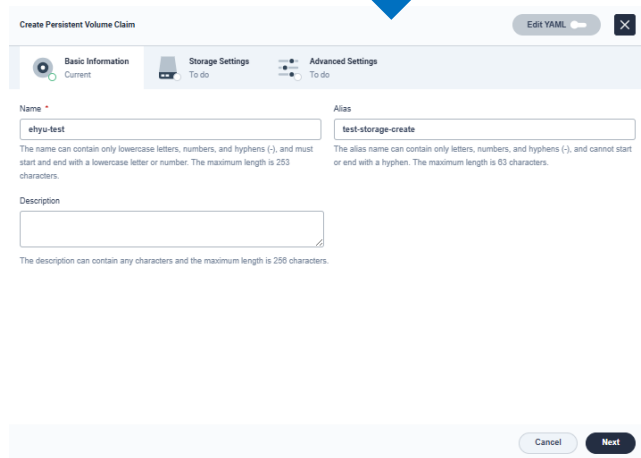
예제1. 스토리지 볼륨 생성

1. 로그인

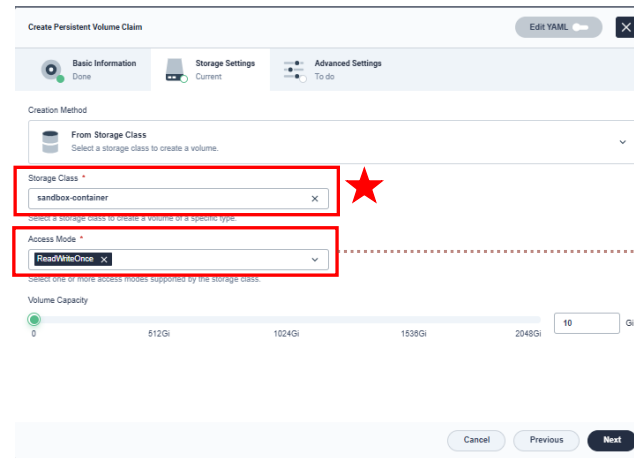
① 'Projects(virl)'>'Storage'>'Persistent Volume Claims'



② 'Create'



③ 정보 입력 후 'Next'



여러 컨테이너에 공유 볼륨 생성 시
ReadWriteMany 추가
ex) 실습 데이터셋용 볼륨

④ 'Storage Class'= 'sandbox-container' 지정

> 'Access Mode' 선택 (원하시는 타입)

> 'Volume Capacity'는 2048Gi를 넘어도 됨. (최대 4TiB)

⑤ 'Create'

Create Persistent Volume Claim

Basic Information Done Storage Settings Done Advanced Settings Current

▼ Add Metadata
Add metadata to the volume.

Cancel Previous **Create**



Edit YAML

```

1 kind: PersistentVolumeClaim
2 apiVersion: v1
3 metadata:
4   name: ehyu-test
5   namespace: vir1
6   creationTimestamp: '2025-06-02T07:31:25Z'
7 annotations:
8   kubernetes.io/alias-name: test-storage-create
9   kubernetes.io/creator: eunho
10  pv.kubernetes.io/bind-completed: 'yes'
11  pv.kubernetes.io/bind-by-controller: 'yes'
12  volume.beta.kubernetes.io/storage-provisioner: rook-ceph.cephfs.csi.ceph.com
13  volume.kubernetes.io/storage-provisioner: rook-ceph.cephfs.csi.ceph.com
14 finalizers:
15   - kubernetes.io/pvc-protection
16 spec:
17   accessModes:
18     - ReadWriteOnce
19   resources:
20     requests:
21       storage: 10Gi
22   volumeName: pvc-2f0d9098-3b6a-4a17-aed5-0672be126ccf
23   storageClassName: sandbox-container
24   volumeMode: Filesystem
25
  
```

※ 별도 yaml로 관리 가능
생성된 볼륨>More>Edit YAML

KUBESPHERE

Persistent Volume Claims

Persistent volume claims define storage requirements. The system creates persistent volumes according to persistent volume claims.

Name	Status	Persistent Volume	Access Mode	Mount Status	Creation Time
ehyu-test(test-storage-create)	Bound	pvc-2f0d9098-3b6a-4a17-aed5-0672be126ccf	RWO	Not mounted	2025-06-02 10:31

Show: 10 Total: 1

⑥ 볼륨 생성 및 상태 확인

>Status ● Bound !상 생성

IV

overview

예제2. 컨테이너 생성

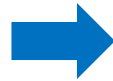
예제에서는 Docker Hub 컨테이너 이미지 'nvidia/cuda:11.6.2-devel-ubuntu20.04' 이미지를 활용하여 리소스를 할당하고, ssh를 설치하여 외부에서 진입하는 것까지 진행함.

- 사전 작업 : 예제1 스토리지 볼륨 생성
- 준비물 : Docker Hub 컨테이너 이미지(사용자 이미지)

① 'Projects(virl)'>'Application Workloads'>'Workloads'>'Create'



② 정보 입력후 'Next'



③ 'Add Container'



④ Docker Hub 이미지 입력

Create Deployment

Basic Information Done Pod Settings Current Storage Settings To do Advanced Settings To do

← Add Container

Container Settings

Set the image, name, type, and computing resources of the container.

Image

Docker Hub nvidia/cuda:11.6.2-devel-ubuntu20.04

To use a private image registry, you need to first create an image registry secret.

nvidia/cuda:11.6.2-devel-ubuntu20.04 Updated 2 years ago No default ports config nvidia Registry

11.6.2-devel-ubuntu20.04

Container Name *

container

Container Type

Worker container

The name can contain only lowercase letters, numbers, and hyphens (-), and must start and end with a lowercase letter or number. The maximum length is 63 characters.

Set the resource limits and requests of the container so that the container is scheduled to appropriate nodes.

CPU Request No request Core CPU Limit No limit Core Memory Request No request Mi Memory Limit No limit Mi GPU Type NVIDIA GPU GPU Limit 8

Available Quotas

Resource Requests: CPU No limit, Memory No limit

Resource Limits: CPU No limit, Memory No limit

GPU Type: nvidia.com/gpu

GPU Limit: 24

Cancel Previous Next



Create Deployment

Basic Information Done Pod Settings Current Storage Settings To do Advanced Settings To do

CPU Request No request Core CPU Limit No limit Core Memory Request No request Mi Memory Limit No limit Mi GPU Type NVIDIA GPU GPU Limit 8

Available Quotas

Resource Requests: CPU No limit, Memory No limit

Resource Limits: CPU No limit, Memory No limit

GPU Type: nvidia.com/gpu

GPU Limit: 24

Port Settings

Set the ports used for accessing the container.

Protocol TCP Name tcp-0 Container Port 22

Add Port

Use Local Image First

Pulls an image only when the required image does not exist locally.

Health Check

Add probes to check the container health status regularly.

Lifecycle Management

Add actions to be performed after the container is started or before it is stopped for environment preparation or graceful shutdown.

Start Command

Customize the command run by the container upon startup. By default, the container runs the default image command.

Command

Startup command of the container

Parameters

Cancel Previous Next

⑤ 이미지 선택 확인

> 'Container Type'은 'Worker container' 고정

> 'GPU Limit' 할당 GPU 수 입력

★ ※ 한 개의 컨테이너에는 최대 8개 까지만 GPU 할당 가능 ★

⑥ 필요 정보 입력 후 '✓' 이후 'Next'

> 'Port Settings', 'TCP', 'Container Port 22'

> 'Start Command' 체크

컨테이너 시작 시 실행할 커맨드 입력 (PID 1 확보)

> 예제에서는 마지막 Edit YAML에서 추가

Create Deployment

Edit YAML

Basic Information Done Pod Settings Done Storage Settings Current Advanced Settings To do

Storage Settings

Mount Volume
Mount an persistent volume, temporary volume, or HostPath volume to the containers.

Mount Configmap or Secret
Mount a configmap or secret to the containers.

Cancel Previous Next



Create Deployment

Edit YAML

Basic Information Done Pod Settings Done Storage Settings Current Advanced Settings To do

Volumes

Mount Volume

Persistent Volume Temporary Volume HostPath Volume

ehyu-test(test-storage-create) 10Gi ReadWriteOnce
sandbox-container Capacity Access mode

container Read and write /data

Cancel Previous Next

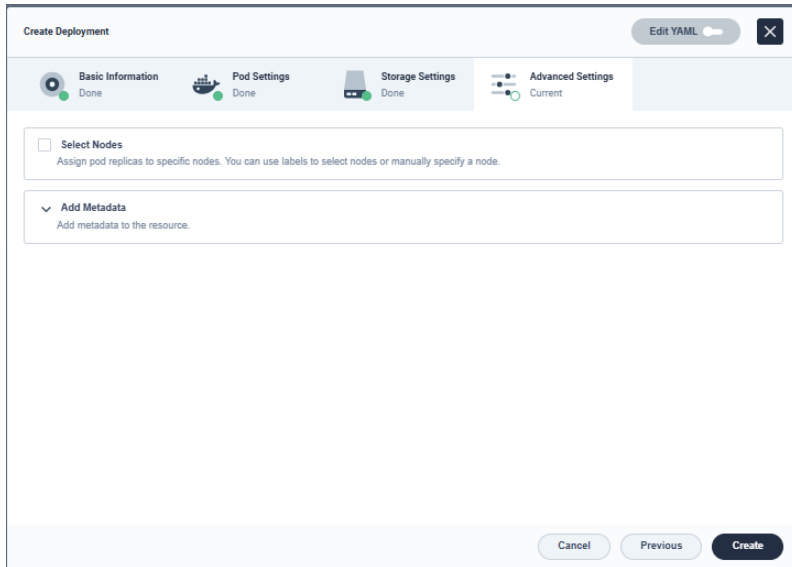
⑦ Storage Settings

- > 'Persistent Volume' > '예제1에서 생성한 스토리지 선택'
- > 'container' > 'Not mounted 원하는 타입 선택'
- > 'Mount Path' 입력

※ 컨테이너 재시작시 마운트한 볼륨을 제외한 데이터 복구 불가

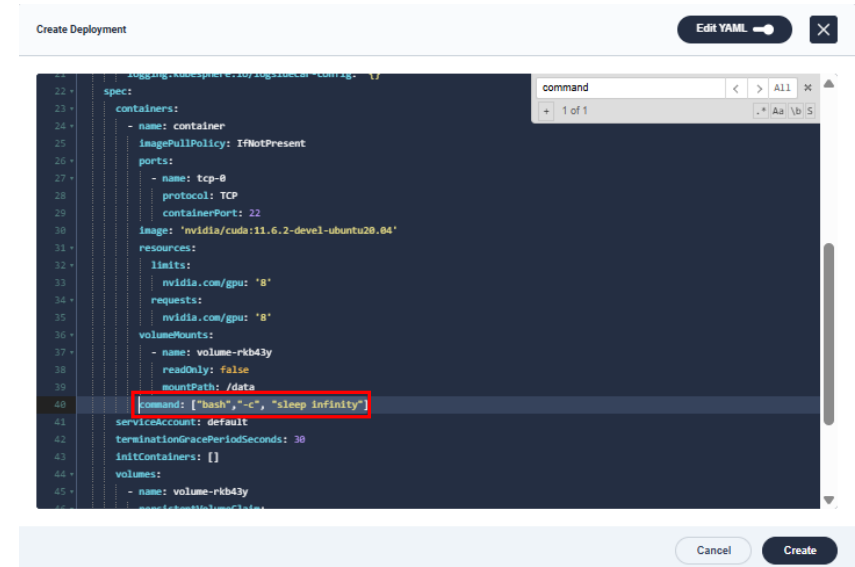
★ 사용자 및 연구 데이터는 해당 볼륨에 저장 ★

⑧ Setting 확인 후 '✓' 후 'Next'



⑧ 'Edit YAML'로 필요 정보 수정

- > GPU를 할당한 시점에서 'Select Nodes' 불필요
- > GPU 스케줄러에 의해 GPU Node 자동 할당

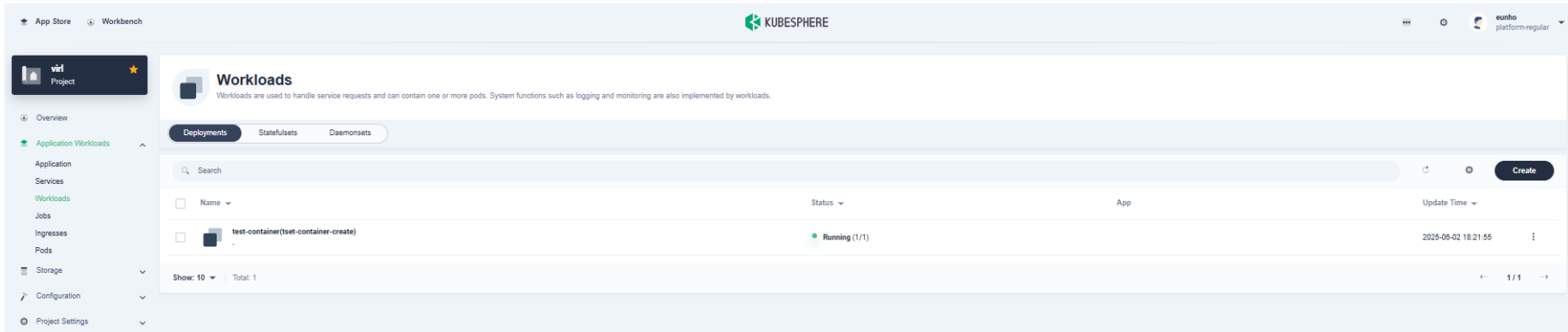


⑨ Edit YAML 작성 후 'Create'

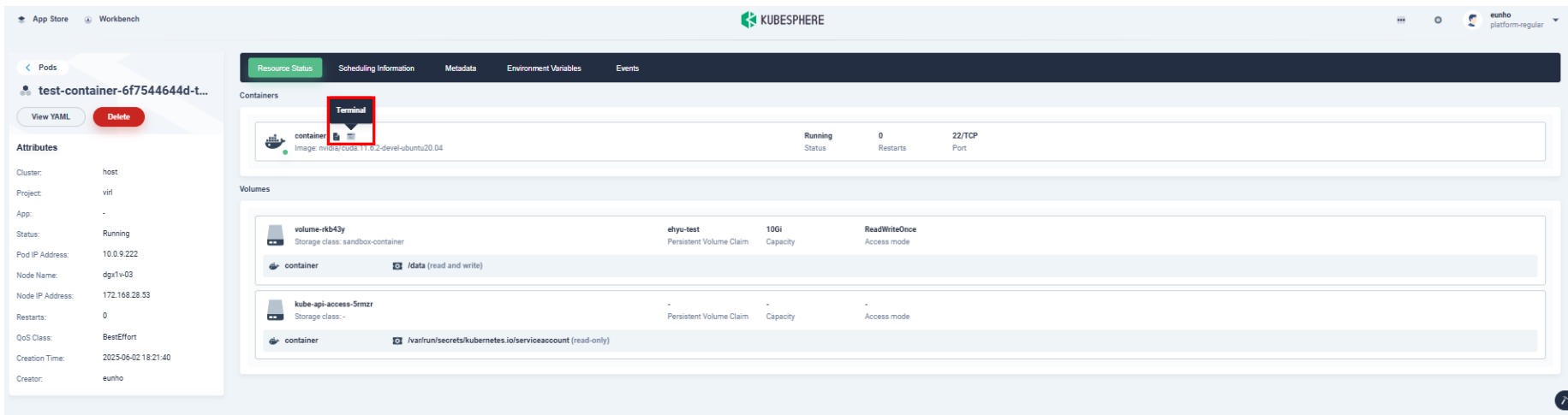
- ⑥에서 체크한 'Start Command' 를 입력하기 위해 'Edit YAMI'
- > .yaml 중에 command: []를 찾은 이후 아래 command 추가
["bash", "-c", "sleep infinity"]
- > PID1번에 bash sleep infinity로 대기 상태로 컨테이너 종료 방지

⑨ 'Workload' 생성 확인 Status가

Running (1/1) 정상 생성 완료



⑩ 생성된 'Workload' 선택 > Pods 컨테이너 선택



⑪ Terminal 아이콘 선택

⑪ 'Storage, GPU' 확인

terminal

```
df -h /data
Filesystem
Size Used Avail Use% Mounted on
nvme0n1 256G 10G 246G 4% /data
```

nvidia-smi

GPU ID	GPU Name	GPU Memory	GPU Utilization	GPU Driver Version	GPU CUDA Version
0	Tesla V100-SXM2-32GB	32GB	0%	450.80	11.8
1	Tesla V100-SXM2-32GB	32GB	0%	450.80	11.8
2	Tesla V100-SXM2-32GB	32GB	0%	450.80	11.8
3	Tesla V100-SXM2-32GB	32GB	0%	450.80	11.8
4	Tesla V100-SXM2-32GB	32GB	0%	450.80	11.8
5	Tesla V100-SXM2-32GB	32GB	0%	450.80	11.8
6	Tesla V100-SXM2-32GB	32GB	0%	450.80	11.8
7	Tesla V100-SXM2-32GB	32GB	0%	450.80	11.8

Processes:

GPU ID	GPU Name	GPU Memory	GPU Utilization
0	Tesla V100-SXM2-32GB	32GB	0%
1	Tesla V100-SXM2-32GB	32GB	0%
2	Tesla V100-SXM2-32GB	32GB	0%
3	Tesla V100-SXM2-32GB	32GB	0%
4	Tesla V100-SXM2-32GB	32GB	0%
5	Tesla V100-SXM2-32GB	32GB	0%
6	Tesla V100-SXM2-32GB	32GB	0%
7	Tesla V100-SXM2-32GB	32GB	0%

\$ df -h /data
\$ nvidia-smi

⑫ 'Terminal' 창이 열리면 작업 진행 (예제에서는 ssh 설치)

terminal

```
Created symlink /etc/systemd/system/multi-user.target.wants/networkd-dispatcher.service.
Setting up openssh-server (1:8.2p1-4ubuntu0.13) ...
Setting up openssh-server (1:8.2p1-4ubuntu0.13) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used.
- at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 76.)
debconf: falling back to frontend: Readline

Creating config file /etc/ssh/sshd_config with new version
Creating SSH2 RSA key; this may take some time ...
3072 SHA256:1X0TE2C1BjAr1U0NMy269F2PvAsr1VktZmK8J0M root@test-container-6f7544644d-twhcn (RSA)
)
Creating SSH2 ECDSA key; this may take some time ...
256 SHA256:40f17g1Mw3BT8X0KadPy13dy20M6572PAK3H0ML0 root@test-container-6f7544644d-twhcn (ECDSA)
)
Creating SSH2 ED25519 key; this may take some time ...
256 SHA256:pyK3309/1Qun/NY1U71ADMBuzthinfv06JAt0tK root@test-container-6f7544644d-twhcn (ED25519)
Created symlink /etc/systemd/system/ssh.service + /lib/systemd/system/ssh.service.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service + /lib/systemd/system/ssh.service.
Inotify: could not determine current runtime
Inotify: policy: denied execution of start.
Setting up ssh-import-id (5.10-ubuntu1.24) ...
Attempting to convert /etc/ssh/ssh_import_id
Processing triggers for systemd (245.4-ubuntu1.24) ...
Processing triggers for libc-bin (2.31-0ubuntu9.12) ...
#
```

\$ apt update
\$ apt install openssh-server -y
\$ mkdir /run/ssh
\$ /usr/sbin/sshd



overview

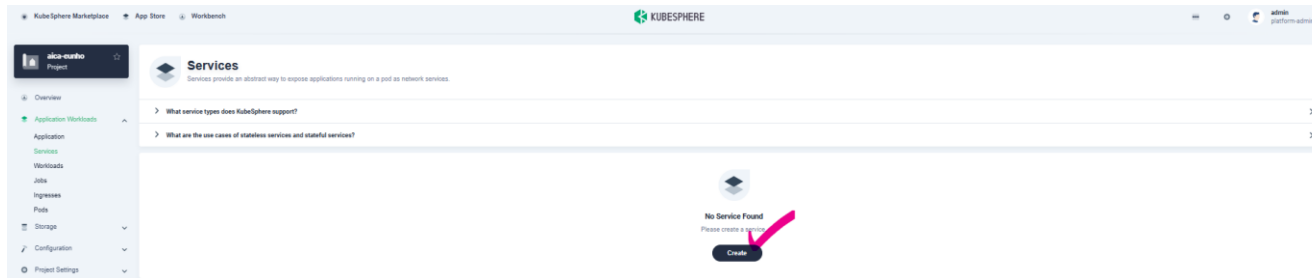
예제3. 컨테이너 서비스 포트 개방



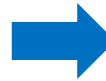
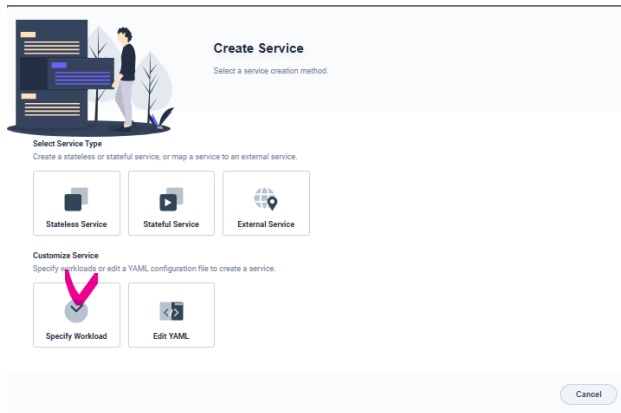
사전 작업 : 컨테이너 생성

① 서비스 생성

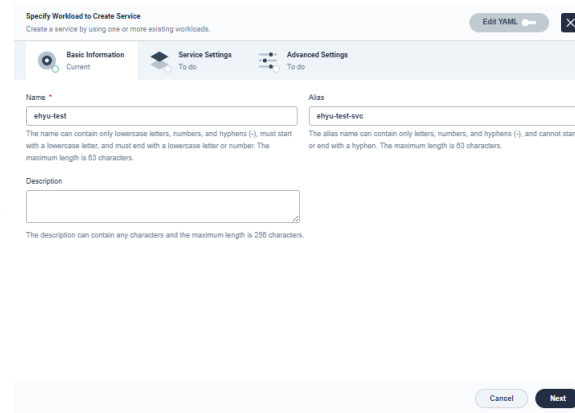
> 'Project'>'Application Workloads'>'Services'>'Create'



> 'Specify Workload' 선택



> 정보 입력 후 'Next'



> 정보 입력 후 'Next'

Specify Workload to Create Service

Create a service by using one or more existing workloads.

Edit YAML

Basic Information Done Service Settings Current Advanced Settings To do

Internal Access Mode

Virtual IP Address ① 고정

Workload Selector *

Specify Workload ← ② 클릭 후 컨테이너 선택

Ports

Set the container ports and service ports.

Protocol HTTP Name http- Container Port Service Port Required Add

③ Protocol / Name / Container Port / Service Port 입력
> Container Port, Service Port는 동일하게 입력
필요시 Add를 통해 포트 추가

Cancel Previous Next

Use labels of a workload as the selector.

Deployments Statefulsets Daemonsets

ehyu-test

Cancel OK

Ports

Set the container ports and service ports.

Protocol TCP Name tcp-ssh Container Port 22 Service Port 22 Add

> 정보 입력 후 'Create'

Specify Workload to Create Service

Create a service by using one or more existing workloads.

Basic Information Done Service Settings Done Advanced Settings Current

☐ External Access ① 체크 'Access Mode'는 'NodePort' 고정
Set the method for accessing the service from outside the cluster.

☐ Session Persistence
Set the system to forward all requests from the same client to the same pod within a specified duration.

^ Add Metadata
Add metadata to the service.

Labels *

app ehyu-test

Add

Cancel Previous ② 생성 Create

☒ External Access
Set the method for accessing the service from outside the cluster.

Access Mode

NodePort

② 서비스 생성 확인 및 접속 포트 확인

> 'External Access' 포트 번호만 확인 (컨테이너 내부 포트 22번: 외부 접속 포트 32570 포워딩 됨)

Name	Service Type	App	Internal Access	External Access	Creation Time
test3	Custom Service VirtualIP	test3	10.99.51.149 VirtualIP	32570/TCP Node ports	2025-08-08 13:46:04

Show: 10 Total: 1

③ 컨테이너로 접근 시에는 210.125.70.71:32570으로 접속

공인 IP는 모든 컨테이너가 동일하며 포트만 구분

부록. 컨테이너 포트 추가 필요 시

① 추가 포트 설정

> 'Project' > 'Application Workloads' > 'Services' > 'Edit Service'

① 선택

② Edit Service

> 'Ports' > 'Add'

> 'Protocol', 'Name', 'Container Port', 'Service Port' 입력 후 OK

Container Port: 443, Service Port: 443



Container Port: 443, Service Port: 443

'Container Port'와 'Service Port'는 동일하게

② 추가 포트 생성 확인

> 'Project' > 'Application Workloads' > 'Services' > 'Edit Service'

Name	Service Type	App	Internal Access	External Access
test3	Custom Service VirtualIP	test3	10.99.51.149 VirtualIP	32570/TCP Node ports

컨테이너 내부 443번 포트에 바인딩 된
외부 포트 32718이 생성된 것을 확인